Elo Rating Methodology

1. Initialization

All teams are initialized with a baseline Elo rating of 1500, which represents the neutral prior strength. As Elo is a relative rating system, absolute values are not meaningful; only differences between ratings influence match probabilities.

$$R_{start} = 1500$$

(Citation: Elo, 1978)

2. Match Prediction

The expected probability of team A winning against team B is computed using the standard Elo logistic function:

$$E(A) = 1 / (1 + 10^{((R_B - R_A) / 400)})$$

This function maps rating differences to win probabilities, ensuring diminishing returns for large rating gaps.

(Citation: Elo, 1978)

3. Update Mechanism

The observed match outcome (S) is encoded as a binary variable (1 for a win, 0 for a loss). The core Elo update rule compares observed performance against expected performance:

$$R_{new} = R_{old} + K * (S - E)$$

(Citation: Elo, 1978)

4. Dynamic K-Factor

To facilitate faster learning for new teams while maintaining stability for established ones, a dynamic K-factor is applied based on the number of matches (M) played:

$$K = K_0 / \sqrt{1 + M/c}$$

where K_0 is the base volatility and c is a scaling constant.

(Citation: Glickman, 1999)

5. Margin of Victory (Dominance)

To quantify dominance, Runs Per Wicket (RPW) is utilized as it unifies run and wicket margins:

```
RPW = Runs / (Wickets + 1)
```

Relative match dominance is computed as the ratio of the winner's scoring efficiency to the loser's. To prevent excessive rating swings from blowout matches, logarithmic scaling is applied:

```
Dominance = RPW_winner / (RPW_loser + epsilon)
MOV = ln(1 + Dominance)
```

The final update equation incorporating dominance is:

```
Delta_R = K * MOV * (S - E)
```

(Citations: Lemmer, 2011; Hvattum & Arntzen, 2010; Silver, 2015)

6. Reliability and Effective Elo

To account for sample size reliability, a confidence function is defined:

```
Confidence(M) = 1 - e^(-M/c)
```

To rank teams accurately, an Effective Elo is computed using mean-centered shrinkage. This dampens the ratings of high-uncertainty teams towards the global average (1500) rather than zero:

```
R_eff = 1500 + (R - 1500) * Confidence(M)
```

(Citations: Glickman, 2012; Herbrich et al., 2007)

7. Temporal Decay

Time-based decay is applied uniformly to normalize ratings over inactive periods (e.g., between seasons), using an exponential half-life model:

$$R_{decayed} = 1500 + (R_{current} - 1500) * e^{(-\ln(2) * t / H)}$$

where H is the half-life parameter and t is the elapsed time. This preserves relative ordering while curbing inflation.

(Citation: Glickman, 2001; FIFA, 2018)

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